

Annotated References on Monitored Fermion Dynamics, Topology, and State Preparation

A research-navigation guide built from the complete bibliographies of Jian–Bauer–Keselman–Ludwig (FGTN) and Bhuiyan–Pan–Jian (BPJ), with a forward literature scan through August 13, 2026. Provenance tags identify works cited by FGTN, BPJ, both papers, or the new scan; preprint and journal versions are counted once.

Community map. The guide follows the conceptual path from monitored-circuit criticality and Gaussian trajectory methods, through localization and dynamical topology, to feedback, state preparation, hardware, and counting theory. Sections I–V concern the dynamics and its classification; VI–IX concern control, preparation, and implementation; Sec. X separates physical continuous records from counting-field book-keeping.

Selected corpus: 100 works, including 36 works identified by the updated scan.

I. MEASUREMENT-INDUCED ENTANGLEMENT AND PURIFICATION TRANSITIONS

Where information survives, purifies, or becomes critical under monitored evolution.

1. Y. Li, X. Chen, and M. P. A. Fisher, “Quantum Zeno Effect and the Many-Body Entanglement Transition,” *Phys. Rev. B* **98**, 205136 (2018). [BOTH] [\[ARXIV\]](#)

Summary. Large-scale Clifford-circuit numerics found a continuous transition between a volume-law phase and a measurement-dominated Zeno phase. The paper is a canonical origin of the hybrid-circuit formulation and of trajectory entanglement as an order parameter.

2. B. Skinner, J. Ruhman, and A. Nahum, “Measurement-Induced Phase Transitions in the Dynamics of Entanglement,” *Phys. Rev. X* **9**, 031009 (2019). [BOTH] [\[ARXIV\]](#)

Summary. Numerics and an exact percolation mapping in a solvable limit established the now-standard volume-law to area-law measurement-induced entanglement transition. It supplies the basic spacetime and minimal-cut picture against which monitored-fermion transitions are compared.

3. Y. Li, X. Chen, and M. P. A. Fisher, “Measurement-Driven Entanglement Transition in Hybrid Quantum Circuits,” *Phys. Rev. B* **100**, 134306 (2019). [BOTH] [\[ARXIV\]](#)

Summary. Clifford-circuit numerics characterized the steady-state and dynamical critical scaling of the measurement-driven entanglement transition. It fixed a widely used benchmark universality class for later Gaussian-fermion studies.

4. C.-M. Jian, Y.-Z. You, R. Vasseur, and A. W. W. Ludwig, “Measurement-Induced Criticality in Random Quantum Circuits,” *Phys. Rev. B* **101**, 104302 (2020). [BOTH] [\[ARXIV\]](#)

Summary. A replica construction maps hybrid-circuit Renyi entropies to a two-dimensional statistical-mechanics model and derives conformal scaling forms at criticality. The work is the field-theoretic template later adapted to Gaussian monitored dynamics and nonstandard replica limits.

5. M. J. Gullans and D. A. Huse, “Dynamical Purification Phase Transition Induced by Quantum Measurements,” *Phys. Rev. X* **10**, 041020 (2020). [BOTH] [\[ARXIV\]](#)

Summary. Theory and circuit numerics identify a purification transition between dynamics that rapidly forget an initially mixed state and dynamics that preserve quantum information for exponentially long times. This operational formulation is especially important for state-preparation protocols, where convergence rather than steady-state entanglement alone is the target.

6. S. Choi, Y. Bao, X.-L. Qi, and E. Altman, “Quantum Error Correction in Scrambling Dynamics and Measurement-Induced Phase Transition,” *Phys. Rev. Lett.* **125**, 030505 (2020). [BOTH] [\[ARXIV\]](#)

Summary. Numerics and quantum-information arguments interpret the entangling phase as a dynamically generated quantum error-correcting code that protects information from local measurements. This viewpoint connects MIPTs to the robustness and resource requirements of monitored state preparation.

7. Y. Bao, S. Choi, and E. Altman, “Theory of the Phase Transition in Random Unitary Circuits with Measurements,” *Phys. Rev. B* **101**, 104301 (2020). [BOTH] [\[ARXIV\]](#)

Summary. Replica statistical mechanics relates the transition to both entanglement and information extraction, with percolation emerging at large local dimension. The paper also makes explicit the equivalence between the entanglement transition and mixed-

free-fermion dynamics.

- 42.** A. W. W. Ludwig, H. Schulz-Baldes, and M. Stolz, “Lyapunov Spectra for All Ten Symmetry Classes of Quasi-One-Dimensional Disordered Systems of Noninteracting Fermions,” *J. Stat. Phys.* **152**, 275–304 (2013). [BOTH] [DOI]

Summary. The paper derives symmetry-enforced structures of Lyapunov spectra for transfer matrices in all ten Altland-Zirnbauer classes. These constraints are the mathematical bridge between monitored single-particle evolution, purification rates, and topological zero or chiral branches.

- 43.** G. Kells, D. Meidan, and A. Romito, “Topological Transitions with Continuously Monitored Free Fermions,” *SciPost Phys.* **14**, 031 (2023). [BPJ] [ARXIV]

Summary. Trajectory numerics show that two competing noncommuting weak-measurement channels stabilize topologically distinct area-law phases and an intervening entanglement regime. A partial-postselection construction links stochastic phases to non-Hermitian topological indices without equating fully monitored dynamics to pure postselection.

- 44.** H. Pan, H. Shapourian, and C.-M. Jian, “Topological Modes in Monitored Quantum Dynamics,” *Phys. Rev. B* **112**, 144301 (2025). [BPJ] [ARXIV]

Summary. Constructed class-DIII and class-A monitored circuits exhibit topological area-law phases whose domain walls retain entanglement against surrounding measurements. Numerical protocols move and braid the DIII dynamical Majorana modes, demonstrating programmable topology in the conditional dynamics.

- 45.** H. Oshima, K. Mochizuki, R. Hamazaki, and Y. Fuji, “Topology and Spectrum in Measurement-Induced Phase Transitions,” *Phys. Rev. Lett.* **134**, 240401 (2025). [BPJ] [ARXIV]

Summary. Numerics show edge-localized Lyapunov zero modes in a topological area-law phase, a bulk Lyapunov gap in trivial and topological localized phases, and bulk gaplessness in the critical phase. A twisted-boundary fermion-parity invariant sharply distinguishes the monitored topological phases.

- 46.** A. H. Bhuiyan, H. Pan, and C.-M. Jian, “Free-Fermion Dynamics with Measurements: Topological Classification and Adaptive Preparation of Topological States,” *Phys. Rev. Research* **8**, 023147 (2026). [BPJ] [ARXIV]

Summary. The paper proves equivalence between many-body-evolution and single-particle-transfer-matrix symmetry classifications for free Gaussian monitored dynamics and formulates their dynamical bulk-boundary correspondence. It also constructs postselection-free adaptive circuits that prepare

and stabilize topological Gaussian states, including constant-depth Chern-state preparation and controllable domain walls.

- 47.** Z. Xiao and K. Kawabata, “Symmetry and Topology of Monitored Quantum Dynamics,” *Phys. Rev. B* **113**, 134307 (2026). [BPJ] [ARXIV]

Summary. A general symmetry analysis classifies Kraus operators and effective non-Hermitian generators and obtains the tenfold topology of monitored free fermions. The paper establishes a dynamical bulk-boundary correspondence in which spacetime topology produces topological steady states and gapless Lyapunov boundary modes, including purification slowdown.

VI. ADAPTIVE FEEDBACK, STEERING, FRUSTRATION, AND ABSORBING DYNAMICS

How records are converted into control actions, dark states, and active or absorbing phases.

- 48.** S. Roy, J. T. Chalker, I. V. Gornyi, and Y. Gefen, “Measurement-induced steering of quantum systems,” *Phys. Rev. Research* **2**, 033347 (2020). [BPJ] [ARXIV]

Summary. This theoretical paper formulates repeated system-detector coupling and detector disposal as a general, postselection-free steering protocol, with few-body and AKLT-chain examples. It is the direct conceptual precursor to viewing measurement backaction plus refreshed ancillas as a state-preparation channel.

- 49.** L. Piroli, Y. Li, R. Vasseur, and A. Nahum, “Triviality of quantum trajectories close to a directed percolation transition,” *Phys. Rev. B* **107**, 224303 (2023). [BPJ] [ARXIV]

Summary. The paper analytically argues, using effective tensor networks, that trajectories generically become disentangled before a product-state absorbing transition is reached. Large stabilizer simulations support the separation and identify the entanglement transition with the feedback-free hybrid-circuit fixed point.

- 50.** P. Sierant and X. Turkeshi, “Controlling entanglement at absorbing state phase transitions in random circuits,” *Phys. Rev. Lett.* **130**, 120402 (2023). [BPJ] [ARXIV]

Summary. Stabilizer-circuit numerics with classical flags show that short- and long-range feedback produce different entanglement scalings at an absorbing transition. The work clarifies when entanglement fluctuations inherit absorbing-state criticality and when the two phenomena remain decoupled.